

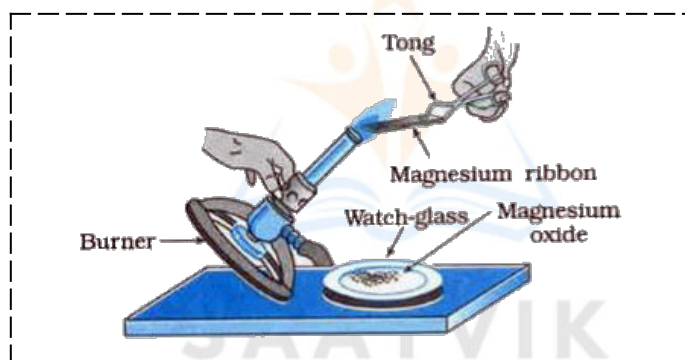


CHEMICAL REACTIONS AND EQUATIONS

Whenever a chemical change occurs, we can say that a chemical reaction has taken place.

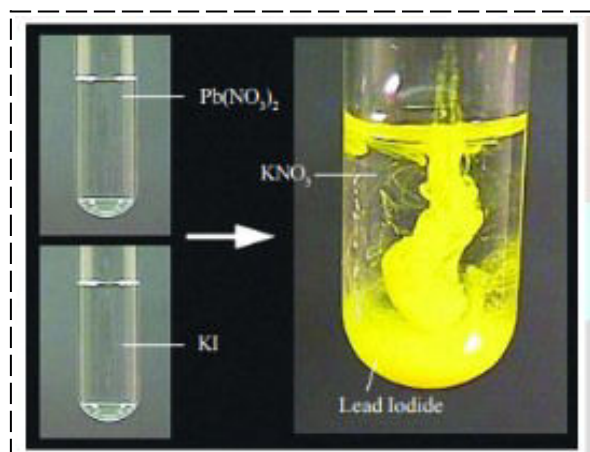
Activity 1.1:

- Magnesium ribbon is cleaned by rubbing it with sandpaper before burning.
- Magnesium reacts with atmospheric oxygen to develop a layer of magnesium oxide. This layer does not allow the underlying magnesium to undergo combustion.
- Magnesium ribbon burns with a dazzling white flame and changes into a white powder. This powder is magnesium oxide.
- It is formed due to the reaction between magnesium and oxygen present in the air.
- $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$



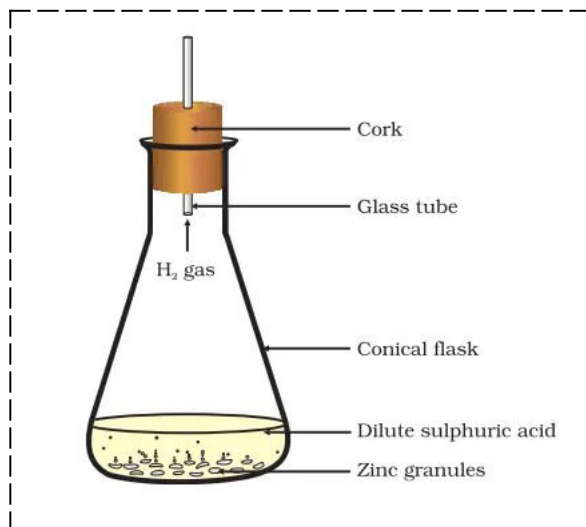
Activity 1.2:

- Lead nitrate reacts with Potassium Iodide to form Lead Iodide (yellow precipitate) and Potassium nitrate.
- $\text{Pb}(\text{NO}_3)_2 + 2\text{KI} \rightarrow \text{PbI}_2 \text{ (yellow ppt)} + 2\text{KNO}_3$



Activity 1.3:

- Formation of hydrogen gas by the action of dilute sulphuric acid on zinc.
- $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$



Chemical Equations

Balanced Chemical Equation:

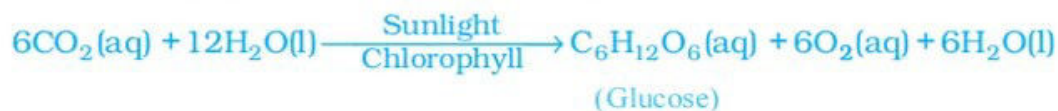
Definition: A balanced chemical equation is an equation in which the number of atoms of various elements is equal on both sides of the equation.

Reason for balancing: An equation should be balanced due to the law of conservation of mass which states that mass can neither be created nor destroyed in a chemical reaction. That is, the total mass of the elements present in the products of a chemical reaction has to be equal to the total mass of the elements present in the reactants.

Writing Symbols of Physical States:

States	Symbol
gaseous	g
liquid	l
aqueous	aq
solid	s

Sometimes the reaction conditions, such as temperature, pressure, catalyst, etc., for the reaction are indicated above and/or below the arrow in the equation. For example:



Types of Chemical Reactions

1. Combination Reaction
2. Decomposition Reaction
3. Displacement Reaction
4. Double Displacement Reaction
5. Oxidation and Reduction

Some other types:

- Exothermic reaction
- Endothermic reaction
- Precipitation reaction
- Redox reaction

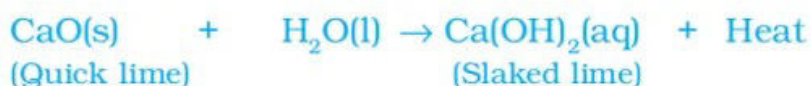
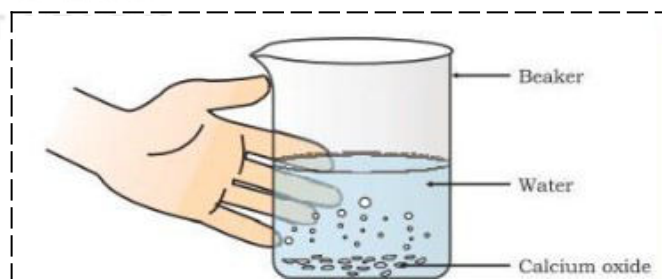
Combination Reaction

Definition:

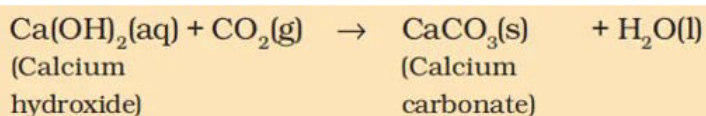
A reaction in which a single product is formed from two or more reactants is known as a combination reaction.

Examples:

- i. Calcium oxide reacts vigorously with water to produce slaked lime (calcium hydroxide) releasing a large amount of heat.



A solution of slaked lime (**calcium hydroxide**) is used for **whitewashing walls**. It reacts slowly with the carbon dioxide in the air to form a thin layer of calcium carbonate on the walls.



- ii. Burning of coal: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- iii. Formation of water from $\text{H}_2\text{(g)}$ and $\text{O}_2\text{(g)}$:

$$2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$$

Exothermic Reaction:

Definition: Reactions in which heat is released along with the formation of products are called exothermic chemical reactions.

Examples:

i. Burning of natural gas: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$

ii. Respiration:

Carbohydrates are broken down to form glucose. This glucose combines with oxygen in the cells of our body and provides energy. The special name of this reaction is respiration.



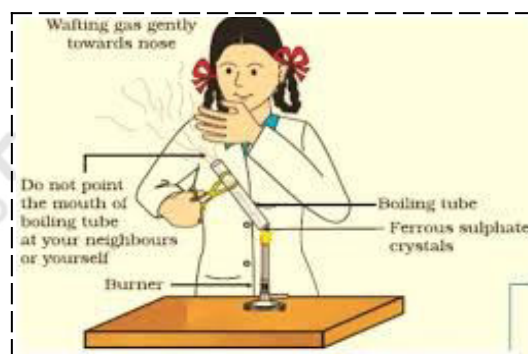
iii. The decomposition of vegetable matter into compost.

Decomposition Reaction

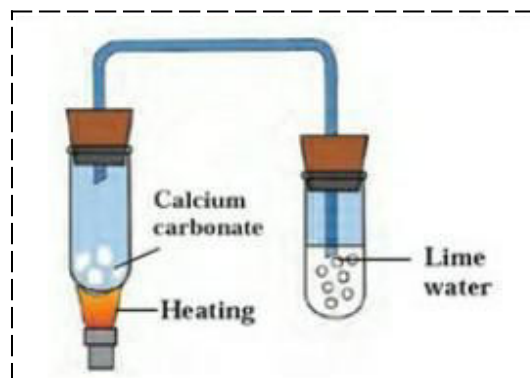
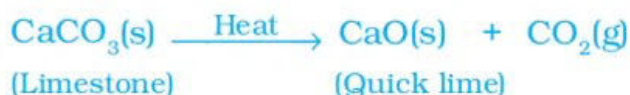
Definition: The reaction in which a single reactant breaks down to give simpler products is called a decomposition reaction.

Examples:

i. Ferrous sulphate crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) lose water when heated and the colour of the crystals changes. It then decomposes to ferric oxide (Fe_2O_3), sulphur dioxide (SO_2), and sulphur trioxide (SO_3). Ferric oxide is a solid, while SO_2 and SO_3 are gases.

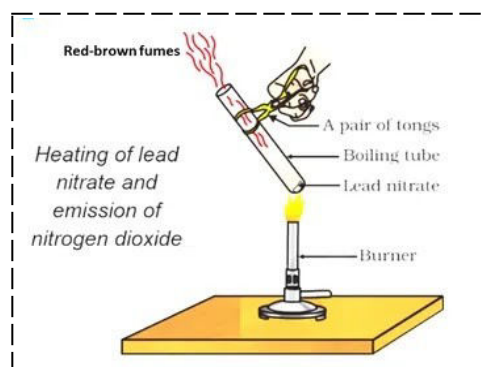


ii. Decomposition of calcium carbonate to calcium oxide (quick lime) and carbon dioxide on heating is an important decomposition reaction.



When a decomposition reaction is carried out by heating, it is called *thermal decomposition*.

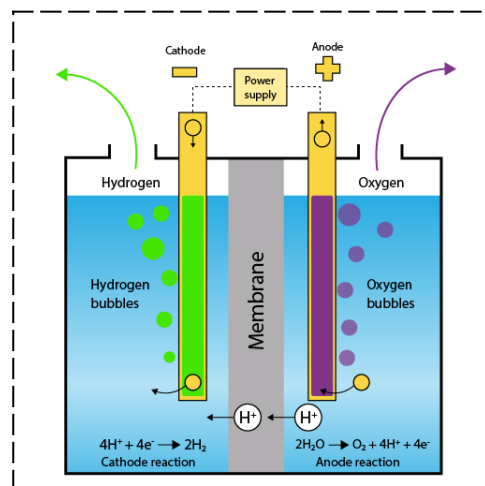
iii. Heating of lead nitrate. You will observe the emission of brown fumes. These fumes are of nitrogen dioxide (NO_2).



iv. **Electrolysis of water** is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during the electrolysis of water is 2:1.

Cathode: Hydrogen; anode: Oxygen.

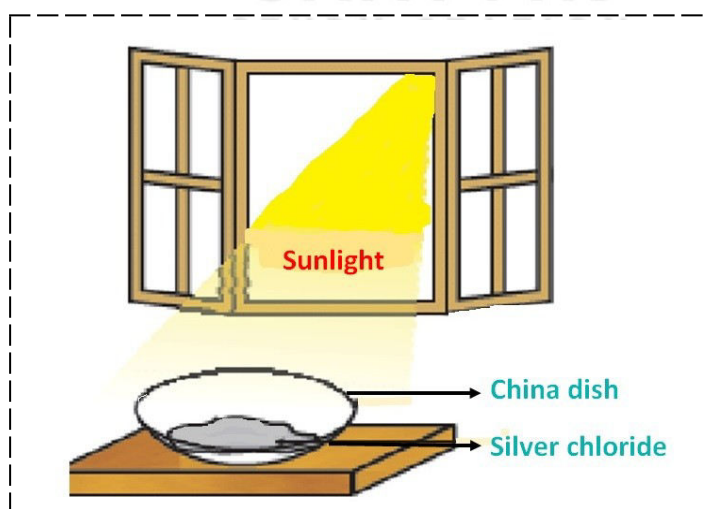
Hydrogen is collected in double the amount of oxygen because water is formed by the chemical combination of hydrogen and oxygen in the ratio 2:1 by volume, so it decomposes in the same ratio.



v. White silver chloride turns grey in sunlight.



The application of this reaction is in **black-and-white photography**.

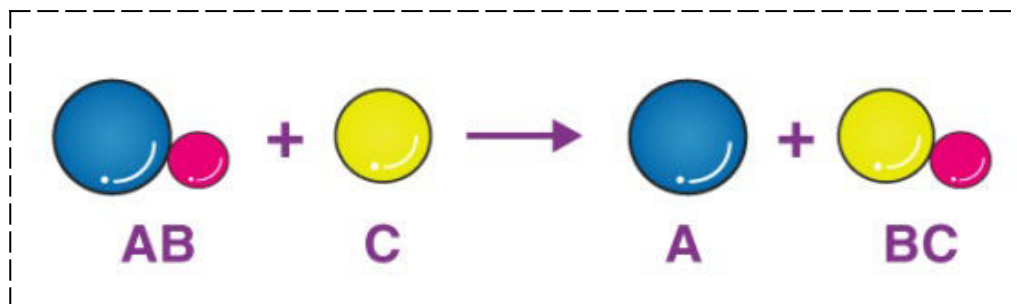


Endothermic reaction:

Reactions in which energy (either in the form of heat, light, or electricity) is absorbed are known as endothermic reactions.

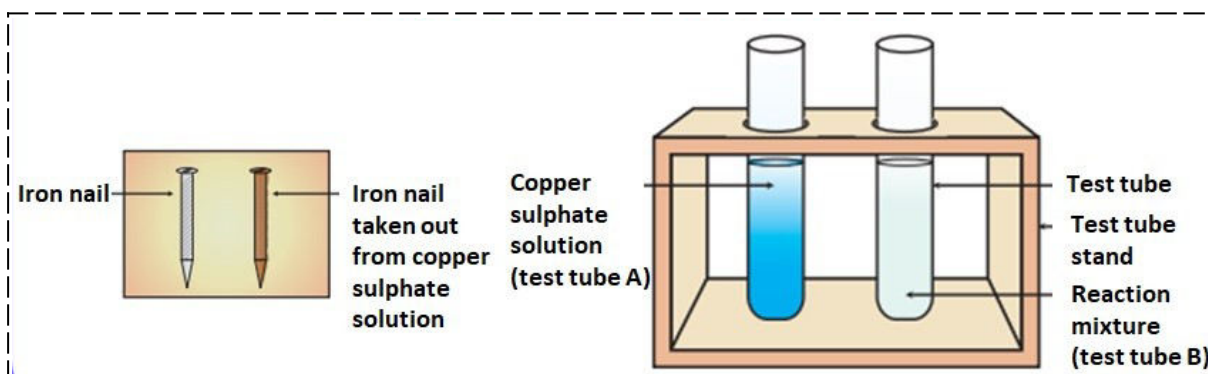
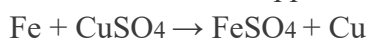
Displacement Reaction

Definition: The reaction in which a more reactive element displaces a less reactive element from its salt solution is called displacement reaction.

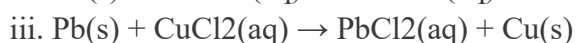
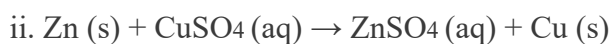


Examples:

i. Iron displaces copper, from copper sulphate solution. Iron nail become brownish in color and the blue colour of the copper sulphate solution fades.

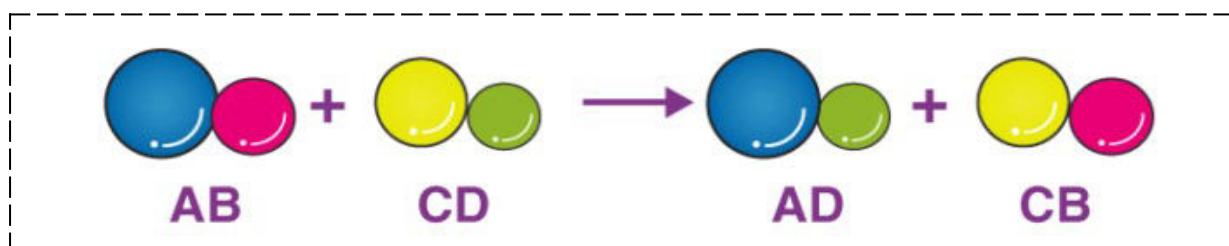


STUDY STATION

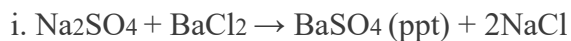


Double Displacement Reaction

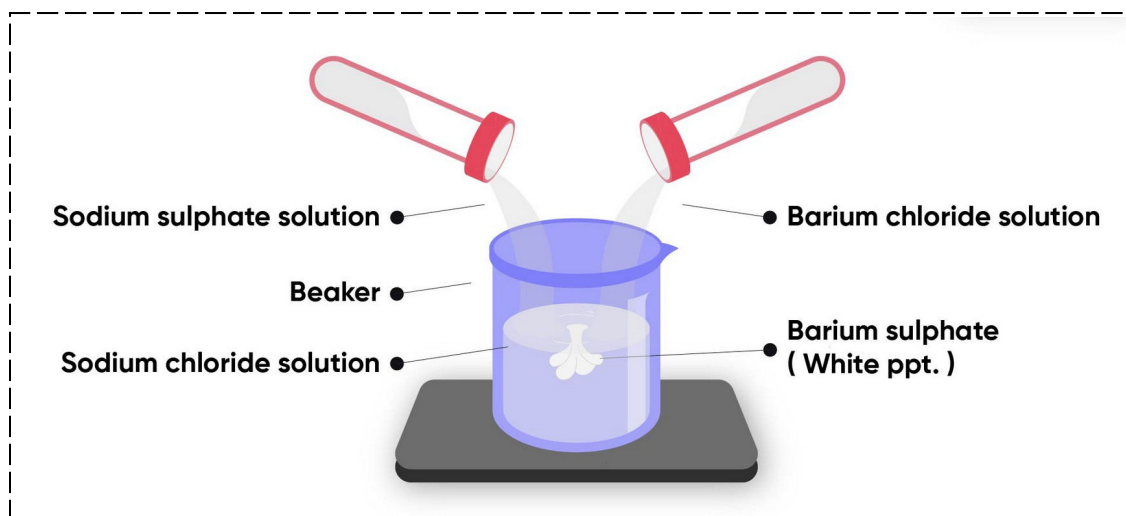
Definition: Reactions in which there is an exchange of ions between the reactants are called double displacement reactions.



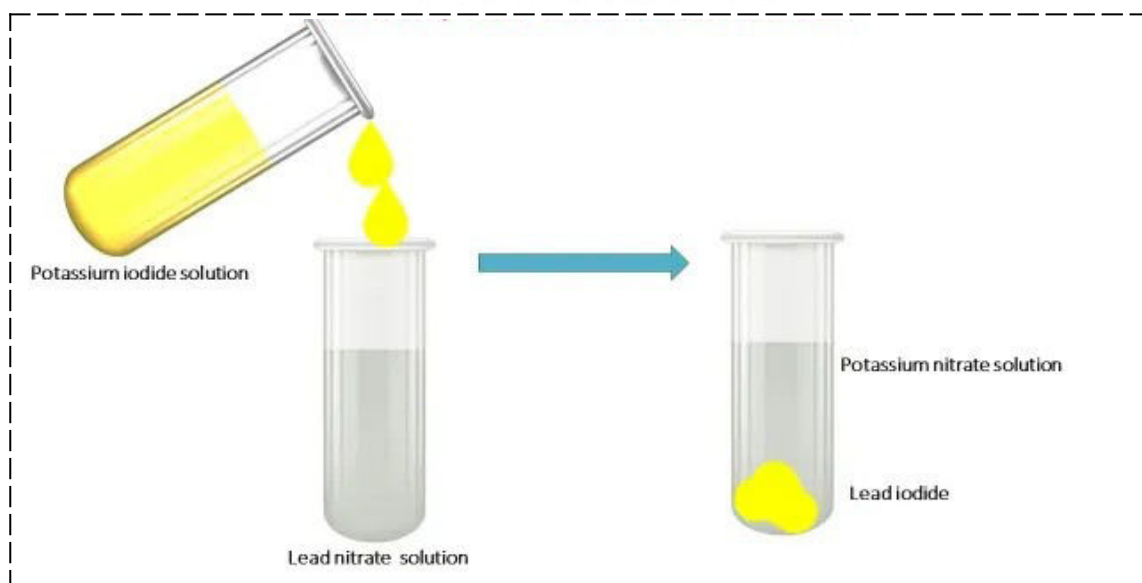
Examples:



You will observe that a white substance (BaSO_4), which is insoluble in water, is formed. This insoluble substance formed is known as a precipitate. Any reaction that produces a precipitate can be called a **precipitation reaction**.

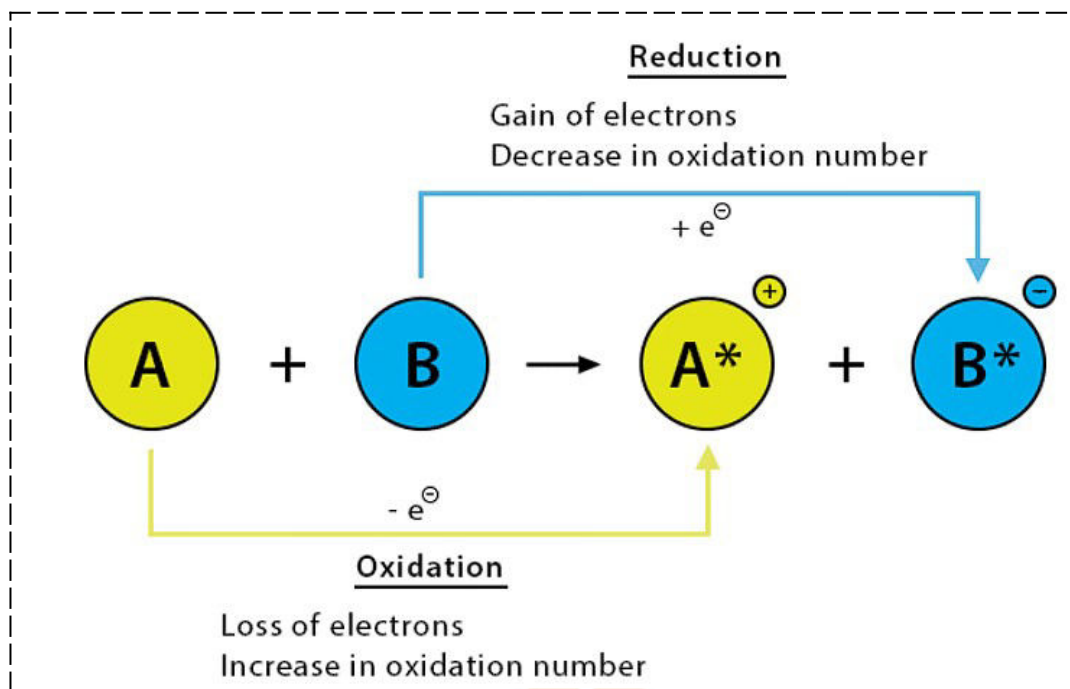


ii. When potassium iodide solution is added to a solution of lead nitrate in a test tube, a yellow color precipitate is formed. The compound precipitated is Lead Iodide.



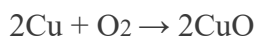
Oxidation and Reduction

Definition: If a substance gains oxygen or loses hydrogen during a reaction, it is said to be oxidized. If a substance loses oxygen or gains hydrogen during a reaction, it is said to be reduced.

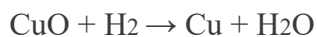


Examples:

i. Oxidation of copper to copper oxide: When we heat copper powder, the surface of copper powder becomes coated with black copper(II) oxide. This is because oxygen is added to copper and copper oxide is formed.



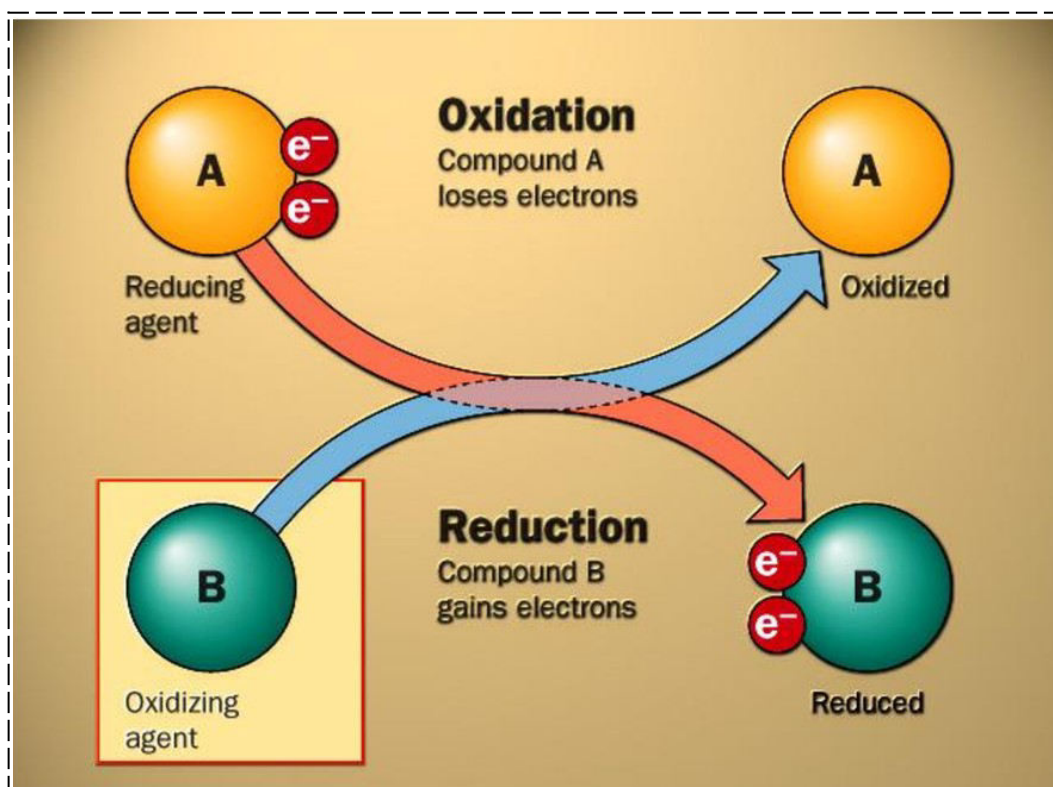
ii. Reduction of copper oxide to copper: If hydrogen gas is passed over this heated material (CuO), the black coating on the surface turns brown as the reverse reaction takes place and copper is obtained.



Redox Reactions

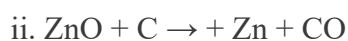
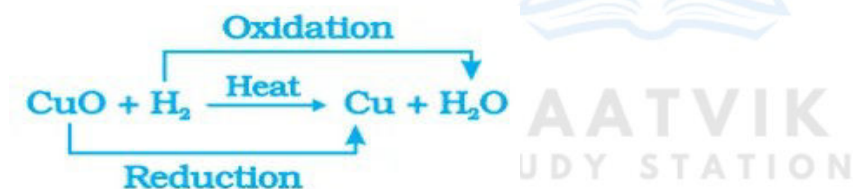
Definition: The reactions in which one reactant gets oxidized while the other gets reduced are called oxidation-reduction reactions or redox reactions.

Note: The substance which gets oxidized is the reducing agent and the substance which get reduced is the oxidising agent.

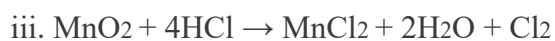


Examples:

i.



Carbon is oxidized to CO and ZnO is reduced to Zn. [Here, Carbon is the reducing agent and ZnO is the Oxidizing agent.]

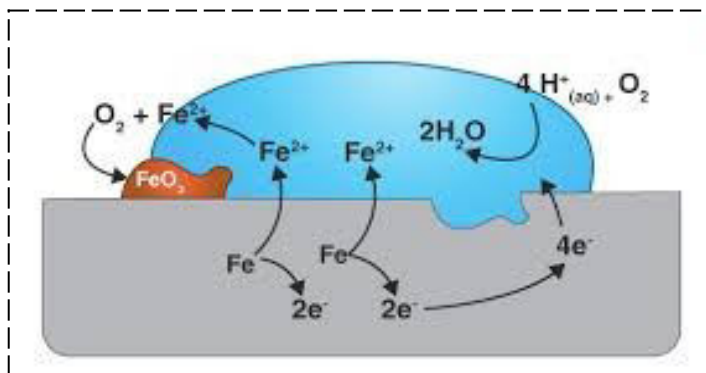


HCl is getting oxidized to Cl_2 while MnO_2 is getting reduced to MnCl_2 . [Here, HCl is the reducing agent and MnO_2 is the oxidizing agent.]

Effects of Oxidation and Reduction in everyday life

i. Corrosion:

When a metal is attacked by substances around it such as moisture, acids, etc., it is said to corrode and this process is called corrosion.



Examples:

- Rusting of iron
- The black coating on silver
- The green coating on copper
-

Disadvantages of Corrosion:

- Corrosion causes damage to car bodies, bridges, iron railings, ships, and to all objects made of metals, especially those of iron.
- Every year an enormous amount of money is spent to replace damaged iron.

ii. Rancidity

When fats and oils are oxidized, they become rancid, and their smell and taste change. Usually, substances that prevent oxidation (antioxidants) are added to foods containing fats and oil. Keeping food in air-tight containers helps to slow down oxidation.

Chips manufacturers usually flush bags of chips with a gas such as nitrogen to prevent the chips from getting oxidized.

